3.10 PN JUNCTION

"When a p-type semiconductor is suitably joined to n-type semiconductor, the contact surface is called pn junction"

Figure 3.26 shows the formation of pn junction. To explain the properties of a pn junction, p-type and n-type semiconductor are suitably joined. Keep in mind that n-type material has a high concentration of free electrons while p-type material has a high concentration of holes. Therefore, at the junction, there is a tendency for the free electrons to diffuse over to the p-side and holes to the n-side.

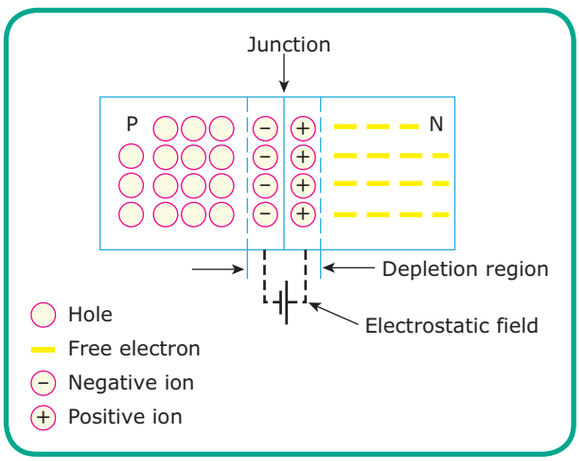


Figure 3.26 PN Junction

This process is called diffusion as shown in Figure 3.27. The combination of these holes and electrons create a new region between the two layers. This region is called "depletion layer". Only inside this, there is positive charge on 'n' side and negative charge on 'p' side. Because of this a potential is produced in this layer which is called "barrier potential". The barrier potential in directly related to depletion layer.

The barrier potential opposes the flow of majority carriers through the junction and it aids the flow of minority carriers. For both of these opposite effects, no charge carriers will flow through the junction at normal condition. The potential difference across a pn junction can be applied in two ways namely, forward biasing and reverse biasing.

3.10.1 Forward Biasing

To apply forward bias, connect positive terminal of the battery to p-type and negative terminal to n-type as shown in Figure 3.28. Due to this, barrier potential is very much reduced. Positive terminal of the battery repels holes in p-side and negative terminal of the battery repels electrons in n-side. Because of this, current flows in the circuit. This is called "forward current". The magnitude of current depends upon the applied forward voltage.

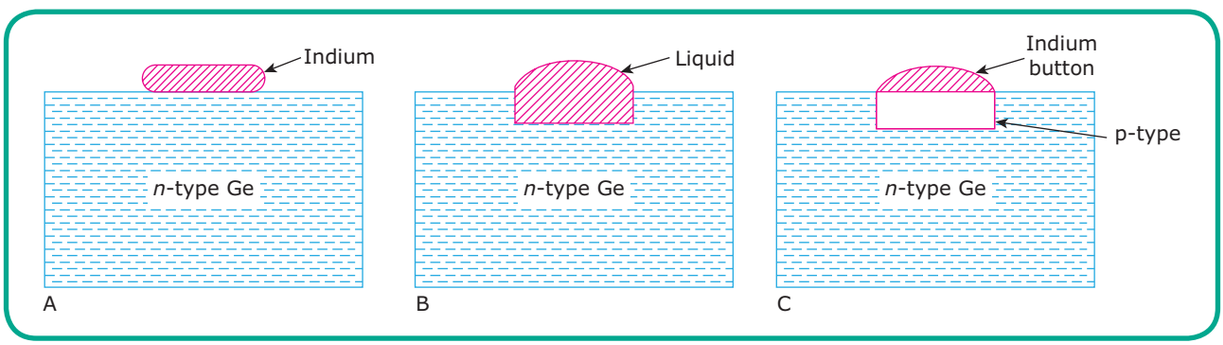


Figure 3.27 Diffusion process

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